

Learning When Black-Hole Hair Is Observable: Physical Identifiability, Generalization, and Observable Complementarity

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Abstract

High interpolation accuracy does not establish that physical parameters are identifiable. We study this distinction in a controlled Kiselev black-hole benchmark with a validated timelike-emitter and three-dimensional null-geodesic shooting calculation. The forward analysis shows that ringdown is informative for the deformation amplitude k , whereas photon geometry supplies an independent response direction that resolves much of the state-parameter w_q degeneracy. Relative to ringdown alone, ringdown plus photon geometry improves the grid-median minimum singular value by $4.54\times$ and the median Jacobian condition number by $2.28\times$. In grouped physical interpolation, the corresponding identifiable- w_q normalized mean absolute error decreases from 0.309 for ringdown to 0.087 for the combined features. Random splitting is more optimistic, while four directional extrapolation tests are substantially harder and split-conformal coverage falls under distribution shift. Photon geometry also improves recovery under a non-learned nearest-model inverse, showing that the gain is not architecture-specific. A uniform 161-phase audit gives a typical normalized prediction shift of 9.48×10^{-4} but exposes catastrophic extrapolation tails. A subsequent targeted 321-phase audit resolves the remaining forward-feature comparisons, yet frozen HGB and random-forest tail shifts exceed predeclared robustness thresholds. Thus the physical complementarity result is supported, but inverse estimator robustness remains conditional. This study is a theoretical identifiability benchmark, not an observational constraint.

1 Introduction

Scientific inverse models can predict accurately for the wrong reason. Repeated realizations of one physical system can leak across random partitions; parameters can be locally degenerate even when observables respond strongly; and a model can interpolate well while failing under extrapolation or small numerical perturbations. These distinctions matter in black-hole spectroscopy, where quasinormal frequencies encode compact-object properties but resolving multiple physical effects remains difficult [8, 2, 3].

Kiselev spacetimes provide a controlled two-parameter setting. They do not stand in for a detector likelihood or a generic beyond-Kerr theory. Their value here is methodological: exact loss of w_q dependence at $k = 0$ supplies a known rank-deficient boundary, while nonzero k permits direct comparison of forward Jacobian geometry and inverse-learning behavior [12].

The study complements rapid gravitational-wave inference based on neural posterior estimation and normalizing flows [7, 11, 10, 6, 17, 16]. Those methods address posterior computation; the

present question is whether the selected forward observables distinguish the parameters in the first place. This distinction parallels structural and practical identifiability analysis in dynamical inverse problems [20, 22].

This work makes four contributions. It (i) evaluates inverse models with leakage-aware random, grouped, and directional protocols; (ii) validates timelike and null Kiselev geodesic shooting with explicit failure accounting; (iii) quantifies physical Jacobian conditioning and observable complementarity; and (iv) audits uncertainty, noise, learning curves, and 81–161–321 phase-resolution robustness under frozen estimators.

The published JCAP study in Ref. [25] supplies the hairy-black-hole and ringdown framework. The present work adds timelike-emitter evolution, three-dimensional null-geodesic shooting, photon-geometry observables, Jacobian identifiability analysis, leakage-aware random, grouped, and directional evaluation, non-learned inverse baselines, numerical-resolution audits, and frozen-estimator robustness tests. This boundary distinguishes the new inverse-learning and physical-shooting study from its theoretical starting point.

The central narrative rests on three distinctions: sensitivity is not identifiability, interpolation is not generalization, and forward convergence is not estimator robustness.

2 Physical Model and Identifiability Framework

The static spherical line element is

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2, \quad (1)$$

$$f(r) = 1 - \frac{2M}{r} - \frac{k}{r^{1+3w_q}}. \quad (2)$$

We use geometrized units with $M = 1$. The dense historical synthetic domain and the later 121-point physical shooting domain are distinct datasets and are not mixed. Leading-eikonal ringdown features use the unstable circular null orbit, with the known limitations of the geodesic-QNM correspondence kept explicit [5, 14, 25].

For standardized observables $\tilde{\mathbf{O}}$ and parameters $\boldsymbol{\theta} = (k, w_q)$, the local Jacobian is

$$J_{ij} = \frac{\partial \tilde{O}_i}{\partial \theta_j}. \quad (3)$$

Higher $\sigma_{\min}(J)$ and lower $\kappa(J) = \sigma_{\max}/\sigma_{\min}$ are favorable. At $k = 0$, $f(r)$ is Schwarzschild for every w_q , so the w_q Jacobian column vanishes exactly. Nonzero rank away from that boundary does not by itself imply practical recoverability at finite precision. The rank-aware historical benchmark that motivated the mature physical calculation is retained in Appendix A and Fig. 13.

Exact $k = 0$ rank loss is structural, while small nonzero k produces practical ill-conditioning at finite precision. The historical proxy experiment suggested that an additional observable family might improve the weakest parameter direction; this motivated replacing those proxies with validated physical shooting observables.

The third panel is therefore historical motivation rather than evidence from the mature physical solver. The physical results below use timelike-emitter and three-dimensional null-geodesic integrations without proxy substitution.

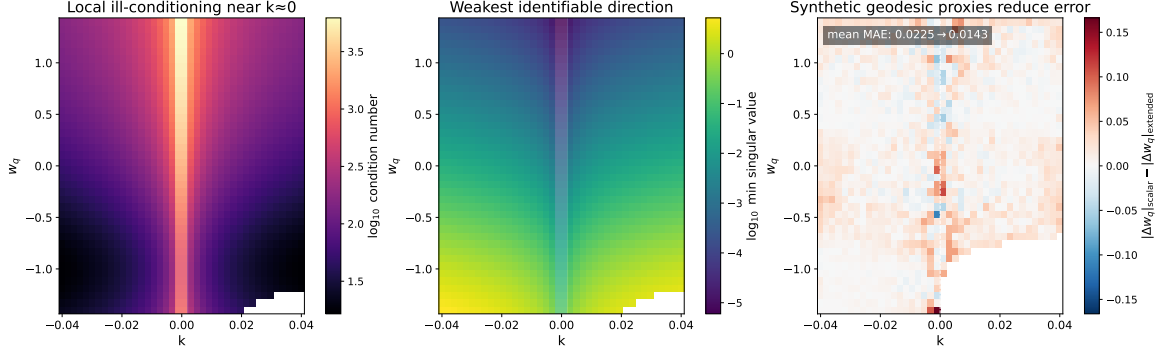


Figure 1: Historical analytic identifiability lens motivating the physical-shooting study. Left: local Jacobian condition number, which grows near $k = 0$. Center: minimum singular value, showing the weakest identifiable direction. Right: change in identifiable- w_q error after adding the historical synthetic geodesic proxies; the mean MAE changes from 0.0225 to 0.0143. White regions are invalid or missing source-grid points and are not interpolated. The first two panels diagnose inverse-map geometry; the third records the original proxy experiment and must not be interpreted as physical ray tracing or geodesic shooting. Proxy augmentation does not resolve the exact $k = 0$ rank loss.

3 Physical Geodesic Shooting and Numerical Validation

The emitter follows the equatorial timelike Hamiltonian

$$H = \frac{1}{2}[-E^2/f + fp_r^2 + L^2/r^2] = -\frac{1}{2}. \quad (4)$$

Its physical coordinate azimuth begins at apocentre: $\phi = \pi$, $r = r_a$, $p_r = t = \tau = 0$. Coordinate ϕ is not treated as a radial anomaly; turning points are detected from $p_r = 0$ and the radial azimuthal period includes apsidal advance.

Photons are propagated in Cartesian spatial coordinates with covariant momentum $\mathbf{p}$ and $p_t = -1$,

$$H_\gamma = \frac{1}{2}[-1/f + \mathbf{p}^2 + (f - 1)(\mathbf{n} \cdot \mathbf{p})^2] = 0. \quad (5)$$

Two launch angles are solved so the direct branch reaches the observer plane. The implementation reports failed phases rather than interpolating them or substituting synthetic geodesic proxies. Photon momenta are projected onto the static observer tetrad before sky angles are formed. Redshift uses $1 + z = \omega_{\text{emit}}/\omega_{\text{obs}}$, and direct arrival time is compared with $\int (1 + z) d\tau_{\text{emit}}$ after removing one common additive constant.

An independently implemented reference solver, which imports no production `bhhairml` physics code, verified the metric and derivative (including explicit M dependence), timelike Hamiltonian and turning-point constants, Cartesian null Hamiltonian and photon initialization, static-observer tetrad, redshift, impact parameter, signed sky coordinates, propagation and arrival timing, physical feature extraction, and selected Jacobians. The maximum trajectory/observable, full feature-vector, and selected-Jacobian absolute differences were $7.633557254893203 \times 10^{-9}$, $7.774758614687016 \times 10^{-11}$, and $2.617905892066119 \times 10^{-13}$, respectively.

In the Schwarzschild benchmark, all 161 phases succeed for both tested nominal w_q values, which agree because $k = 0$. The independent radial benchmark validates the turning-point period and apsidal advance. The physical 121-point grid also preserves the Hamiltonian constraints and exact rank-loss boundary. The overlapping arrival curves and their numerical residuals are retained as supporting evidence in Appendix I and Fig. 15.

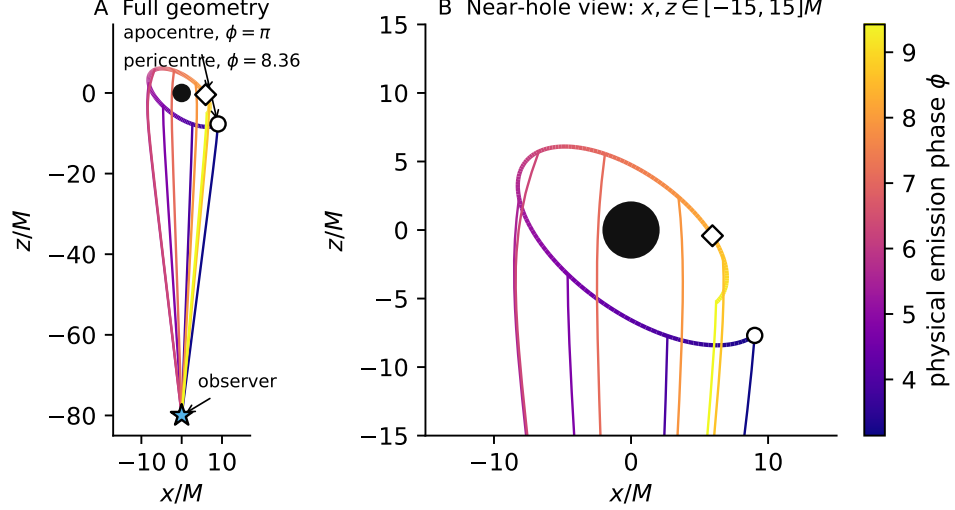


Figure 2: Representative direct-branch null geodesics for $M = 1$, $r_p = 8M$, $r_a = 12M$, $k = 10^{-3}$, $w_q = -0.5$, and observer $(0, 0, -80M)$. Physical phase runs from π at apocentre to 3π ; the diamond marks the numerically detected pericentre and the dark disk is a horizon-scale marker, not a shadow. At each phase, two launch angles are solved until the ray reaches the observer. The right panel shows the stated near-hole coordinate bounds on their actual scale; no trajectory is geometrically magnified. This is a controlled dimensionless benchmark.

Figure 2 makes the direct-branch geometry explicit using actual validated emitter states, launch angles, photon integrations, and observer geometry. It is a controlled benchmark visualization, not a radiative-transfer or detector simulation.

4 Physical Observable Complementarity

A large response can reveal that the spacetime changed without revealing which parameter caused the change. Timing and redshift responses are often large but nearly collinear in parameter space. Photon geometry rotates the sensitivity direction and therefore contributes information that ringdown lacks. Figure 3 first shows how the same three validated spacetimes produce distinct phase-resolved response patterns.

The standardized local comparison is shown in Fig. 4.

The full observer tracks nearly overlap, as expected for these small deformations. Figure 5 therefore retains the physical track as the primary object and shows unnormalized, pointwise matched-phase residuals relative to Schwarzschild. These small resolved changes supply a direction that is difficult to see on the full track.

Across the accepted physical grid, ringdown plus photon geometry improves the median $\sigma_{\min}$ by $4.5429\times$ and the median condition number by $2.2819\times$ relative to ringdown alone. The medians for the combined set are $\sigma_{\min} = 1.2326$ and $\kappa(J) = 4.0731$. Ringdown plus all shooting has $\sigma_{\min} = 1.5312$ and $\kappa(J) = 4.4974$: adding more responsive summaries can raise $\sigma_{\min}$ without producing the best conditioning. Figures 6 and 7 show the pointwise maps and their grid-wide summaries, respectively.

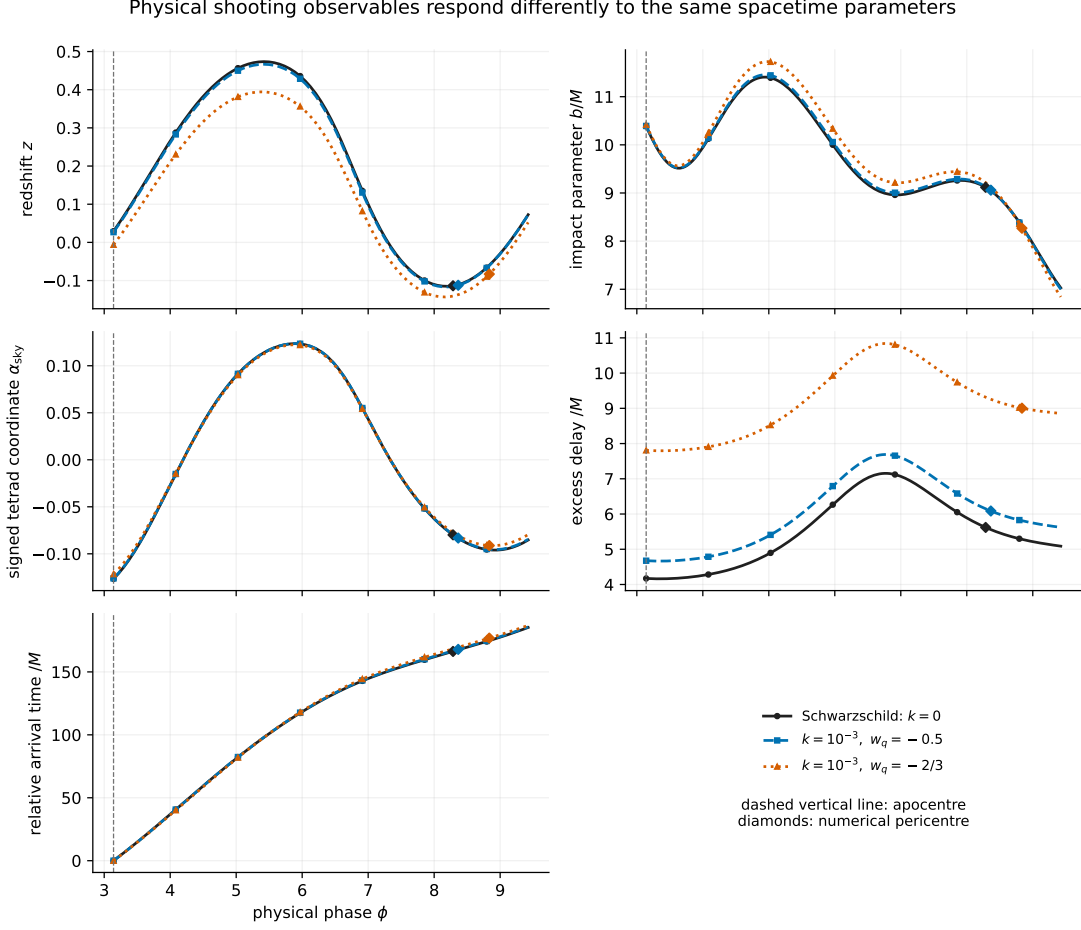


Figure 3: Physical direct-branch observables over $\phi \in [\pi, 3\pi]$ for Schwarzschild and $k = 10^{-3}$ with $w_q = -0.5, -2/3$, using $M = 1$, $r_p = 8M$, $r_a = 12M$, and observer $(0, 0, -80M)$. The dashed line marks the initial apocentre and diamonds mark numerical pericentres. All curves use the same successful matched phases without interpolation. Observable response amplitude and independent parameter information are not equivalent.

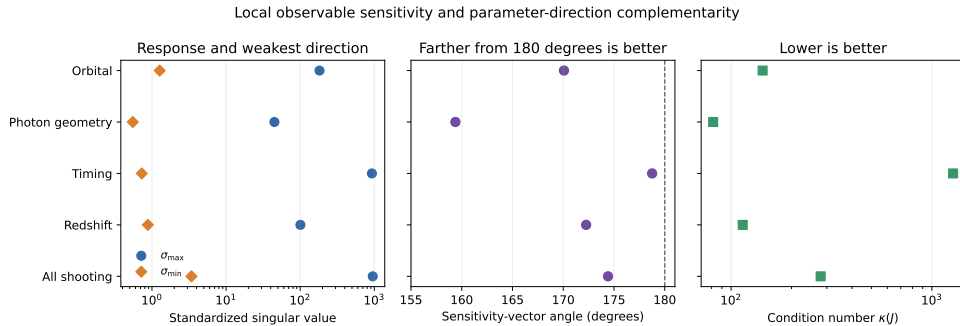


Figure 4: Local standardized response and separation diagnostics. Timing has a large response but an angle of 178.76° and condition number 1275.5; photon geometry rotates the response to 159.37° and lowers the condition number to 81.4. Higher $\sigma_{\min}$, angles farther from 180° , and lower condition number are favorable; magnitude alone is not identifiability.

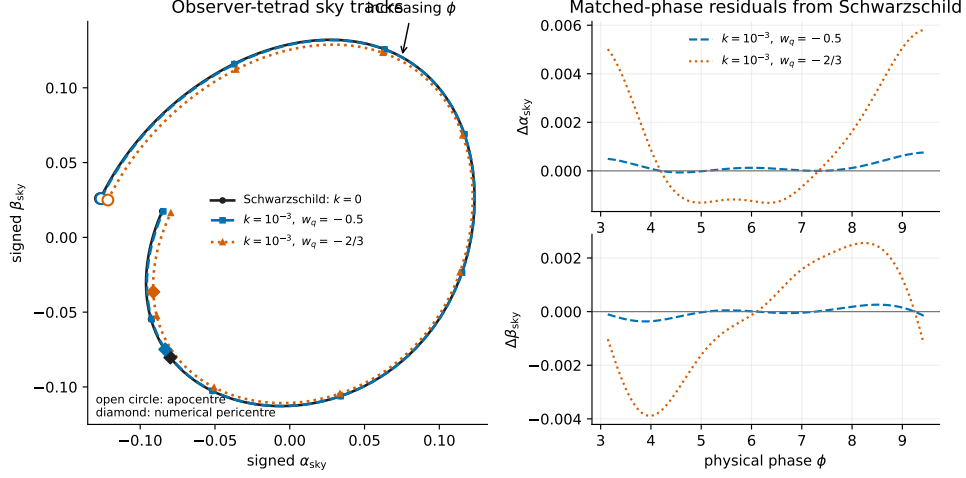


Figure 5: Static-observer-tetrad sky tracks and actual-scale residuals for the same $M = 1$, $r_p = 8M$, $r_a = 12M$, observer $(0, 0, -80M)$ direct-branch benchmark. Open circles mark $\phi = \pi$ apocentre, diamonds mark numerical pericentre, and the arrow gives increasing phase. Residuals are pointwise differences from Schwarzschild at exactly matched successful phases; they are neither normalized for visibility nor interpolated across failures.

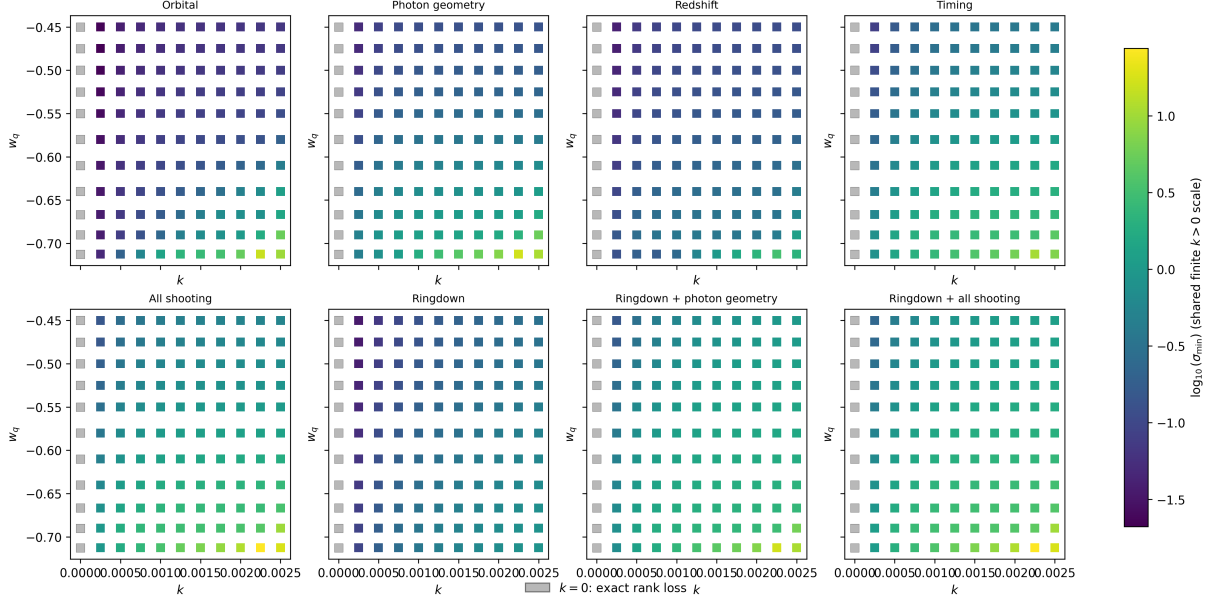


Figure 6: Grid maps of $\log_{10} \sigma_{\min}$ with exact $k = 0$ rank-loss points masked. Missing or invalid values are not interpolated, and all panels use one finite- k color scale.

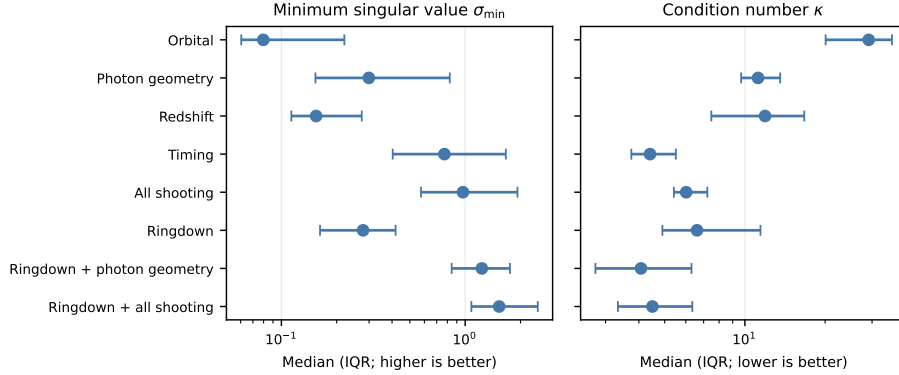


Figure 7: Grid-wide median and interquartile summaries of $\sigma_{\min}$ and condition number. Photon geometry provides the clearest complement to ringdown.

Table 1: Primary physical-complementarity summaries. The reported combined-set gains are the archived grid-wide improvement statistics relative to ringdown, not ratios formed from separately reported medians.

Features	med. $\sigma_{\min}$	med. $\kappa(J)$	gain
Ringdown	0.2782	6.6112	reference
Rdown + photon geom.	1.2326	4.0731	$4.54 \times / 2.28 \times$
Rdown + all shooting	1.5312	4.4974	—

4.1 Finite-domain ambiguity check

Because local Jacobian rank does not establish global injectivity, we also tested the sampled physical domain for distant observable near-collisions. Across all 121 accepted physical points, no finite- k system had a nearest observable neighbor farther than $d_\theta = 0.25$ in normalized parameter space, and no pair with $d_\theta \geq 0.25$ fell below the fifth-percentile local observable spacing. For ringdown plus photon geometry, the closest pair with $d_\theta \geq 0.5$ has RMS standardized observable distance 0.03155. This is approximately 12.8 times the largest measured 161-to-321 combined-feature RMS forward-resolution change, 0.002462.

At $d_\theta \geq 0.5$, photon geometry reduces the number of distant pairs below local-median observable spacing from 13 to 6, increases the closest-pair distance from 0.02517 to 0.03155, and decreases the maximum nearest-observable-neighbor parameter distance from 0.23216 to 0.21886. It therefore reduces the most distant sampled ambiguities. It does not preserve every local nearest-neighbor ordering and rearranges some nearby systems in observable space, even while improving the more consequential distant sampled separation. Full finite-grid results and limitations appear in Appendix M.

5 Leakage-Aware Inverse-Learning Protocols

Five primary feature sets are ordered consistently: ringdown, photon geometry, all shooting, ringdown plus photon geometry, and ringdown plus all shooting. Here photon geometry and the other integrated quantities are physical shooting observables, distinct from the historical synthetic geodesic proxies. Models are histogram gradient boosting (HGB), random forest (RF), and a target-scaled multilayer perceptron (MLP) [4, 9, 18]. These heterogeneous standard estimators are retained intentionally rather than optimizing one high-capacity architecture: tree models expose piecewise split-crossing behavior, while the MLP supplies a smoother perturbation response and

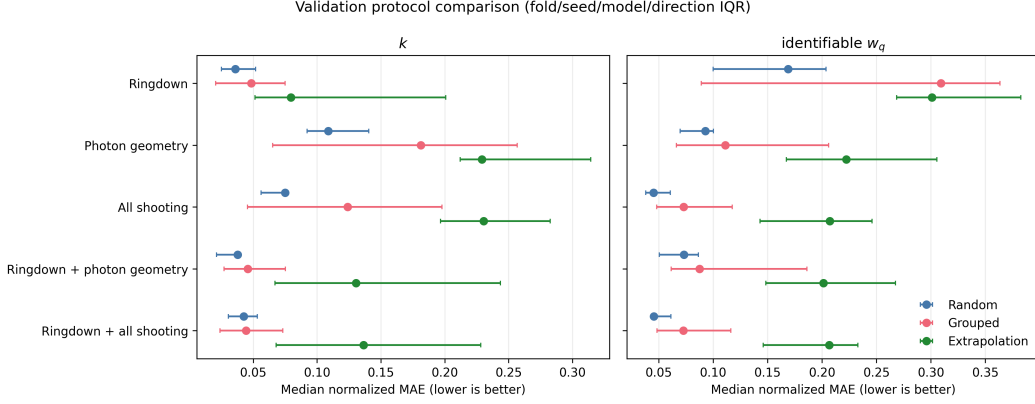


Figure 8: Median NMAE for the five primary feature sets under random, grouped physical interpolation, and directional extrapolation. Directions remain separate in the machine-readable source before aggregation.

a distinct extrapolation failure mode. Their purpose is to reveal estimator-specific behavior, not to claim that standard models are universally preferable. The protocols are random interpolation, two blocked 3×3 grouped physical layouts, and four directional extrapolation tests. All nominal $k = 0$ records are retained and flagged, but ordinary w_q error excludes them because w_q is exactly non-identifiable there.

Normalized mean absolute error divides MAE by the full target span. Split-conformal intervals target 90% coverage and are interpreted under their exchangeability assumptions [23, 21, 1]. Feature-noise scales and every preprocessing transform are learned on training folds. The 81-to-161 and targeted 161-to-321 comparisons apply archived frozen estimators; they do not retrain on higher-resolution test features.

6 Inverse-Learning Results

Random splitting is optimistic. For HGB, RF, and MLP considered together, random ringdown NMAE is 0.0361/0.1688 for k /identifiable w_q , while grouped ringdown is 0.0485/0.3093. Ringdown plus photon geometry gives grouped 0.0457/0.0875, with the w_q improvement reproduced independently in both tree families. Directional extrapolation remains harder: the combined set has aggregate NMAE 0.1305/0.2012. Figure 8 compares the three validation regimes directly.

6.1 Traditional inverse baselines

To separate physical complementarity from architecture-specific learning, we evaluate two non-learned inversions under exactly the same registered physical holdouts. The nearest physical model fits a standardization transform on the training observables only, computes every Euclidean distance to the training catalog explicitly, and returns the (k, w_q) pair of the nearest training system. The local Jacobian inverse selects its reference from the training catalog, constructs a training-only local linear forward map, and applies the Moore–Penrose pseudoinverse. Calibration and test systems participate in neither construction.

For the nearest baseline, identifiable- w_q NMAE decreases from 0.123810 to 0.086580 under random interpolation, from 0.220833 to 0.148148 under grouped interpolation, and from 0.254161 to 0.195026 under directional extrapolation when photon geometry is added to ringdown. These

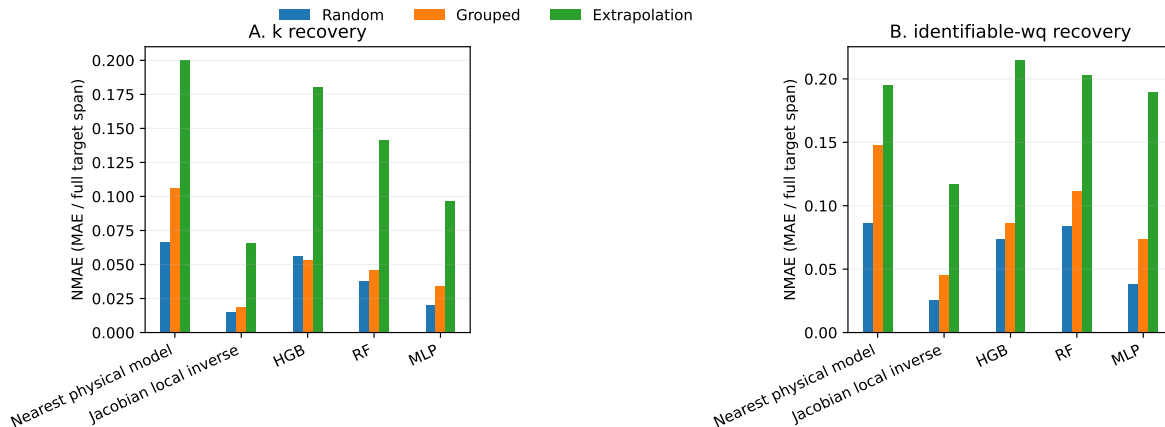


Figure 9: Traditional and learned inverse methods using ringdown plus photon geometry. Bars show median NMAE for k and identifiable w_q under random, grouped, and directional protocols. The nearest and Jacobian baselines are deterministic conditional on each registered split, whereas archived ML summaries aggregate the existing folds and seeds. The Jacobian bars must be read with the unclipped out-of-domain diagnostic reported in the text.

are reductions of 30.1%, 32.9%, and 23.3%, respectively. Observable complementarity is therefore a property of the physical forward map rather than merely an effect learned by a particular estimator; nearest-neighbor inversion is not asserted to be optimal.

For ringdown plus photon geometry, the local Jacobian median k /identifiable- w_q NMAE is 0.0147/0.0259, 0.0184/0.0454, and 0.0661/0.1172 for random, grouped, and directional protocols. However, 1315 of 9950 raw predictions lie outside the registered physical domain and were not clipped. Thus strong local inversion performance coexists with poor global robustness; favorable median local-linear performance does not establish a reliable global inverse. The full method comparison is shown in Fig. 9, with detailed values and conditioning associations retained in Appendix E.

Conditioning and coverage explain different aspects of error. In the error model, standardized training distance has estimate 0.330, extrapolation status 0.276, and log condition number 0.197. Physical conditioning generally agrees with grouped tree-based w_q performance, but it does not perfectly rank every model, target, or protocol. The MLP supplies a counterexample: it can have reasonable central metrics while producing catastrophic directional tails. The traditional-versus-learned comparison appears in Fig. 9; the conditioning associations are retained separately in Appendix E.

Nominal 90% conformal coverage is 0.892/0.965 for random k/w_q , but falls to 0.722/0.851 under grouped interpolation and 0.509/0.660 under extrapolation. This is empirical coverage under shift, not a retained finite-sample guarantee. Figure 10 separates coverage by target and protocol.

7 Numerical Convergence and Estimator Robustness

The uniform 161-phase study completes all 121 physical points. From 9317 feature-point comparisons, 240 are unresolved under the frozen criterion; the median normalized feature change is 7.908×10^{-4} and the maximum is 0.08069. Across frozen scored predictions, the median normalized shift is 9.482×10^{-4} and the 95th percentile is 0.02606. The median aggregate NMAE change is 6.880×10^{-4} , but the maximum is 16.8497 because of unclipped MLP directional extrapolation

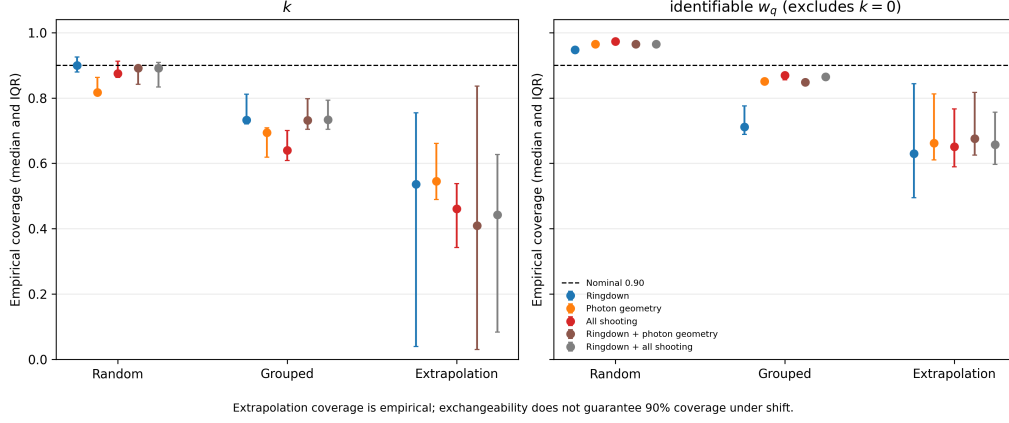


Figure 10: Empirical 90% split-conformal coverage by target and protocol. Ordinary w_q coverage excludes $k = 0$; extrapolation coverage is not guaranteed under distribution shift.

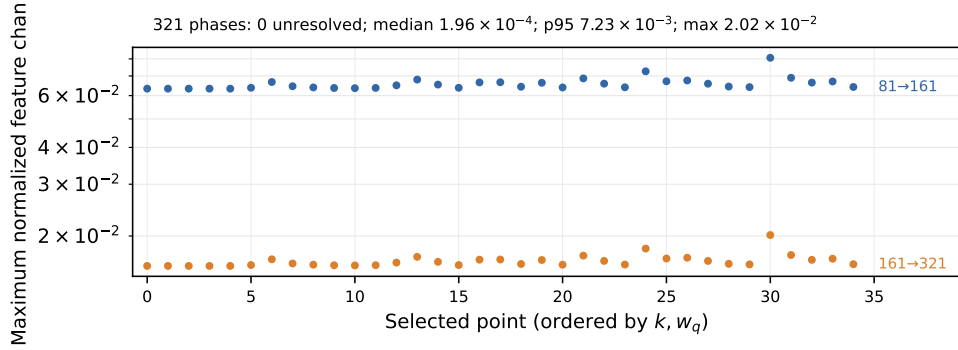


Figure 11: Targeted pointwise maximum normalized feature changes from 81 to 161 and 161 to 321 phases. The 321-phase audit resolves the forward-feature criterion but does not by itself certify inverse estimators.

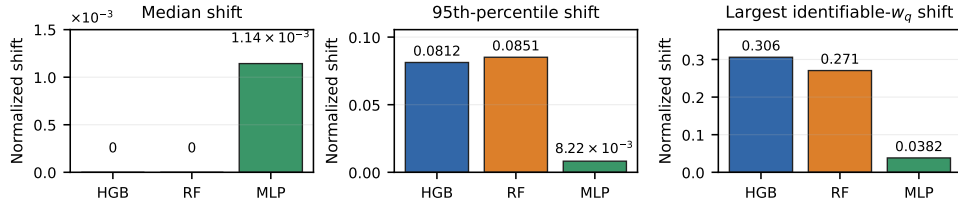
outliers.

The targeted 321-phase audit selects 35 points deterministically before reading their 321-phase outcomes. All complete. Median and 95th-percentile normalized 161-to-321 feature changes are 1.963×10^{-4} and 7.234×10^{-3} ; no comparison remains unresolved, ringdown is exactly unchanged, and Jacobian rank is preserved. Forward convergence therefore passes. Figure 11 displays both resolution increments on the same selected-point order.

Frozen trees remain tail-sensitive: HGB and RF 95th-percentile normalized shifts are 0.08122 and 0.08509, and their largest identifiable- w_q shifts are 0.30593 and 0.27053. Their median shifts are zero because many tree predictions do not cross a split threshold. The predeclared perturbation ensembles and fixed Extra Trees smoother do not pass all stability and grouped-performance criteria. Of 410 catastrophic MLP records, 409 are already catastrophic at 161 phases, none is materially worsened by the 321-phase substitution, 63 reach the iteration limit, and no numerical overflow is detected. The decision is Outcome B: the forward map converges, but estimator robustness does not.

Typical predictions are stable, but the tails are not. The earlier MLP maxima are not representative of every estimator, while the targeted tree failures show that the remaining issue also cannot be attributed solely to the MLP. The inverse-estimator robustness conclusions therefore remain conditional.

Estimator sensitivity after 161-to-321 forward convergence



MLP audit: 410 catastrophic records; 409 predate phase substitution; 63 iteration-limit cases.

Figure 12: Frozen-estimator sensitivity after forward convergence. Bars show the median, 95th-percentile, and largest identifiable- w_q normalized prediction shift for HGB, RF, and MLP. The audit note summarizes the distinct MLP failure audit. No predictions are clipped.

Figure 12 separates the estimator mechanisms. Tree medians remain exactly unchanged because most perturbations do not cross a split, yet their tails do. The MLP reacts more smoothly to phase substitution, while its catastrophic directional records overwhelmingly predate that substitution.

8 Discussion

To distinguish improved physical observability from architecture-specific learning effects, we also evaluate non-learned nearest-model and local-linear inverse baselines under the same physical hold-out protocols. The nearest-model improvement confirms an estimator-independent geometric contribution. Changes in local conditioning, however, do not consistently predict pointwise recovery across all models and protocols: Spearman associations have mixed signs and broad bootstrap intervals. Jacobian conditioning diagnoses local physical distinguishability, but it is not a complete predictor of finite-data error or behavior under distribution shift, and no causal relationship is inferred.

Jacobian conditioning answers whether nearby parameter directions are separated by the chosen observables. Training distance answers whether a fitted estimator has support where it is evaluated. Neither statistic subsumes the other. Consequently, $\sigma_{\min}$ can predict broad observable-set ordering without ranking every empirical model or protocol.

Tree ensembles are piecewise and can show zero median response followed by large split-crossing tails. The MLP is smoother under the targeted feature substitution but extrapolates to physically unreasonable values in archived directional tests. These mechanisms are estimator-specific manifestations of a general scientific-ML risk: a converged forward simulator does not ensure a stable inverse surrogate [15, 13, 19].

What is specific to Kiselev is the metric deformation and exact $k = 0$ loss of w_q . What may generalize is the workflow: establish forward rank, seek complementary observables, partition by physical systems, test directional shift, and perturb frozen estimators at independently measured numerical scales.

The Jacobian analysis is local by construction. The finite-grid collision audit provides a complementary nonlocal check across the sampled parameter domain: no unresolved distant finite- k collision is found at the tested resolution, although photon geometry rearranges local neighbor ordering. Together with the improved nearest-model recovery, this shows that improved distant sampled separation need not imply preservation of every local observable-space neighborhood.

9 Limitations

The ringdown model is leading-eikonal rather than a dedicated finite- ℓ perturbative spectrum. The spacetime is static. Shooting includes only the direct photon branch, a point emitter and point observer, and no radiative transfer, detector response, secondary images, strain likelihood, calibration error, or model discrepancy. The physical grid is finite. Historical proxy results remain synthetic. Although the targeted audit resolves higher-harmonic forward differences on its selected set, it is not a uniform 321-phase grid and tree tail sensitivity persists. MLP extrapolation is unstable. The verdict is therefore conditional, and no result is an observational constraint. The finite-domain ambiguity audit does not prove continuous global injectivity; the 121-point grid can miss isolated or continuous degeneracies between sampled parameter values.

10 Conclusion

Physical complementarity is supported: ringdown constrains k , while photon geometry supplies an independent direction needed for much of the w_q recovery. Observable complementarity persists under a non-learned inverse baseline, while the local Jacobian inverse demonstrates that strong local recoverability can coexist with globally unstable predictions in this Kiselev benchmark. Random interpolation overstates recoverability relative to grouped and directional validation. Finally, numerical and distribution-shift audits are necessary before confident inverse claims: the forward features can converge while fitted estimators remain fragile. These conclusions motivate robust inverse methods and broader physical modeling before observational use. Within the sampled physical domain, no unresolved distant finite- k observable collision is found at the tested numerical resolution, although this finite-grid result is not a proof of continuous global identifiability.

Data Availability

The analysis code and tracked machine-readable derived results, registered split assignments, and reproducibility audits are available in the project repository [24]. This work uses simulated theoretical data and no observational data. Large intermediate numerical products that are not tracked in the repository are available from the author upon reasonable request.

A Historical Synthetic and Proxy Benchmark

The original dense benchmark used leading-eikonal curves, training-fold PCA, and synthetic geodesic proxies. Waveform-only RF achieved $R^2(k) = 0.8960$ and $R^2(w_q) = 0.9587$; HGB with proxy augmentation achieved 0.9793 and 0.9891. The rank-aware classifier reached macro F1 0.9928 with a 57.6% weak-identifiability fraction. PCA compressed correlated information; it did not create a new physical response direction. These historical results motivated the physical shooting study and are not substituted for it. Figure 13 serves a distinct role from the historical lens in Fig. 1: it compares a learned rank-aware class with the analytic weak-identifiability boundary rather than displaying conditioning or proxy regression-error changes.

B Full Split Definitions

The archived assignment table contains 115 registered split specifications over five seeds. For each seed it contains one random interpolation split, 18 grouped splits (nine held-out blocks for each of

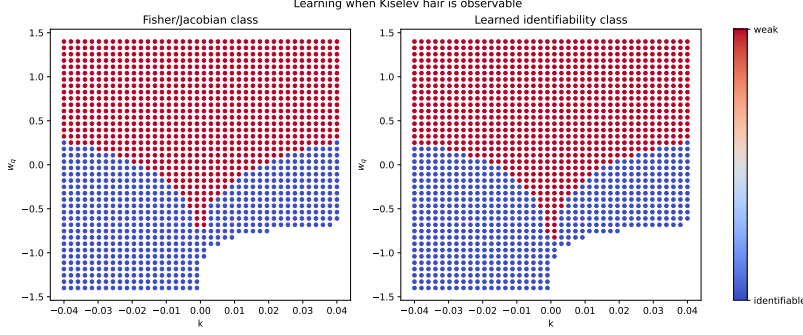


Figure 13: Historical dense synthetic rank-aware identifiability boundary. Exact $k = 0$ rank loss is structural, whereas the finite-width weak-identifiability region depends on precision. This historical benchmark motivated, but does not replace, the validated physical shooting analysis.

two 3×3 layouts), and four directional splits. Random splitting holds out 20% of the 121 physical systems for test and assigns 20% of the remainder to calibration. The grouped layouts use index shifts $(0, 0)$ and $(1, 1)$ before forming the nine contiguous blocks; each block is held out once and 20% of the remaining systems calibrate the fit. Directional tests separately hold out the highest three k levels, lowest three k levels, lowest three w_q levels, and highest three w_q levels, with 20% of the complementary pool used for calibration. Training, calibration, and test systems are disjoint. All $k = 0$ systems remain registered; they contribute to k scoring but not ordinary w_q scoring because w_q is unidentifiable there. Every scaler, noise scale, feature transform, and model is fitted on training data only. No new split was generated for this manuscript.

C Feature Definitions

Table 2 lists the registered feature families. Except for the two orbital invariants and four ringdown scalars, each phase-resolved quantity is represented by its mean, amplitude, reference value where defined, and sine/cosine coefficients through the third harmonic.

Table 2: Registered physical feature families and exact dimensions.

Family	Physical quantities	Source	Dim.
Ringdown	Δr , photon-sphere radius, Ω , Lyapunov exponent	static metric/eikonal model	4
Orbital	radial azimuthal period, apsidal advance; summaries of r and p_r	timelike emitter	20
Photon geometry	summaries of impact parameter and signed sky coordinates	direct null rays	27
Redshift	summaries of photon redshift	emitter/ray/observer	9
Timing	summaries of excess delay and relative arrival time	direct null rays	17
All shooting	orbital, photon geometry, redshift, and timing	physical shooting	73
Ringdown + photon geometry	union of the corresponding families	combined	31
Ringdown + all shooting	union of the corresponding families	combined	77

D Additional Model-Level Results

HGB and RF reproduce the central grouped photon-geometry improvement for identifiable w_q . The MLP is retained as a stress test rather than omitted: its catastrophic errors are un-clipped and included in aggregate summaries.

E Traditional Inverse Baselines

Table 3 gives the complete ringdown-plus-photon-geometry comparison. Each entry is median k /identifiable- w_q NMAE. The nearest and local-linear methods are deterministic for a fixed registered split; their rows are summarized over the same split assignments, while archived learned model rows retain the existing fold/seed aggregation.

Table 3: Traditional and learned inversion with ringdown plus photon geometry. NMAE is MAE divided by the full target span; $k = 0$ remains in k scoring but is excluded from identifiable- w_q scoring.

Method	Random	Grouped	Directional
Nearest physical model	0.0667/0.0866	0.1063/0.1481	0.2000/0.1950
Local Jacobian inverse	0.0147/0.0259	0.0184/0.0454	0.0661/0.1172
HGB	0.0560/0.0740	0.0530/0.0865	0.1803/0.2146
RF	0.0378/0.0844	0.0457/0.1113	0.1418/0.2033
MLP	0.0206/0.0383	0.0344/0.0738	0.0966/0.1896

The primary Jacobian predictions are raw and unclipped: 1315 of 9950 fall outside $0 \leq k \leq 0.0025$ or $-0.7125 \leq w_q \leq -0.45$. Reference condition number, reference $\sigma_{\min}$, and local observable-space distance are stored for every prediction. Figure 14 reports the supported pointwise/fold-level conditioning analysis. Most bootstrap intervals span zero, and the significant nearest-model associations are negative; these mixed results reinforce that local conditioning is not a complete estimator- error model.

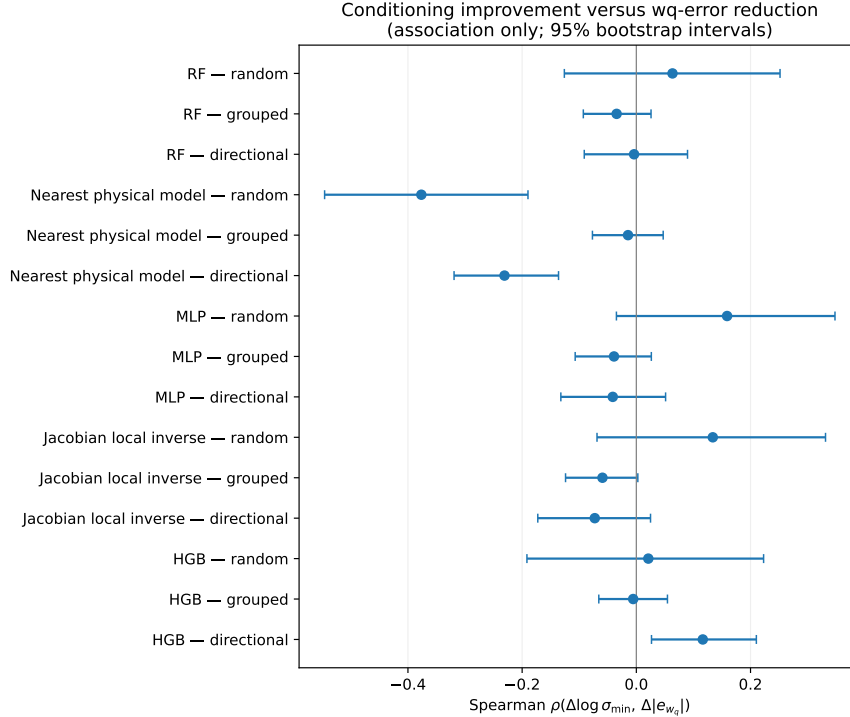


Figure 14: Spearman association between the change in log minimum singular value and the reduction in absolute identifiable- w_q error when photon geometry is added to ringdown. Points show method/protocol estimates and bars are 95% bootstrap intervals. Associations are descriptive and do not imply causation.

F Four Directional Extrapolation Tests

Directions are stored independently in the prediction tables before any median summary. High-to-low and low-to-high tests differ in training support and cannot be interpreted as interchangeable folds.

G Conformal Calibration and Rejection

Split-conformal residual quantiles are fitted on calibration subsets. The exchangeability guarantee applies to matching distributions; shifted coverage is reported empirically. Rejection curves rank uncertainty but do not repair distribution shift.

H Noise and Learning Curves

Grouped one-percent feature noise yields ringdown-plus-photon-geometry NMAE 0.0699/0.0853 for k/w_q . Learning curves are still improving at 78 training points, so data limitation remains part of the interpretation.

I Numerical-Validation Thresholds

The 35-point 321-phase run takes 5318.2 s. Maximum hit, timelike-constraint, null-constraint, and impact-parameter drift errors are respectively 7.0913×10^{-8} , 3.1303×10^{-13} , 7.6214×10^{-12} , and 1.7932×10^{-11} . The Schwarzschild arrival-time comparison in Fig. 15 separates the visually overlapping curves from their actual-scale residuals.

Matched Schwarzschild and Kiselev direct rays are compared on their actual coordinate scale in Fig. 16.

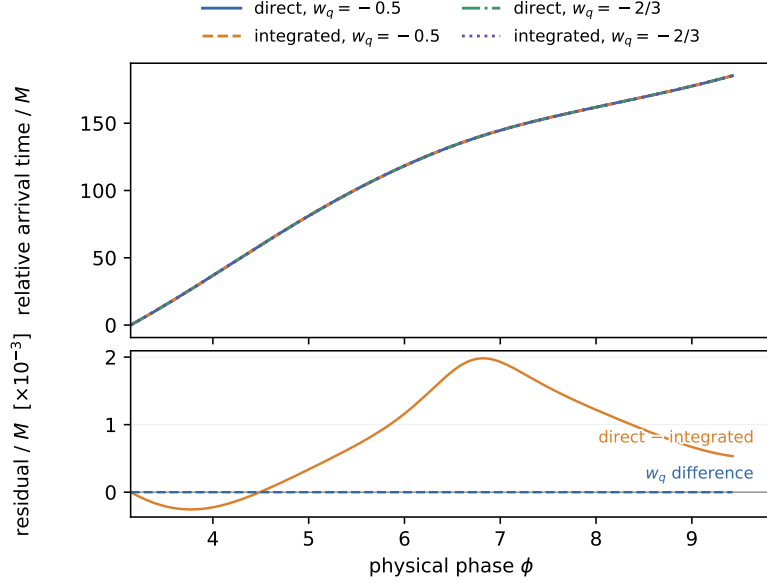


Figure 15: Schwarzschild arrival-time validation. Four arrival-time curves are plotted. The two nominal w_q cases coincide because the $k = 0$ geometry is Schwarzschild, while the direct and integrated-redshift constructions agree within numerical quadrature error. The lower panel shows direct minus integrated arrival time for $w_q = -0.5$ and the direct-arrival difference between $w_q = -0.5$ and $w_q = -2/3$ on their actual numerical scales.

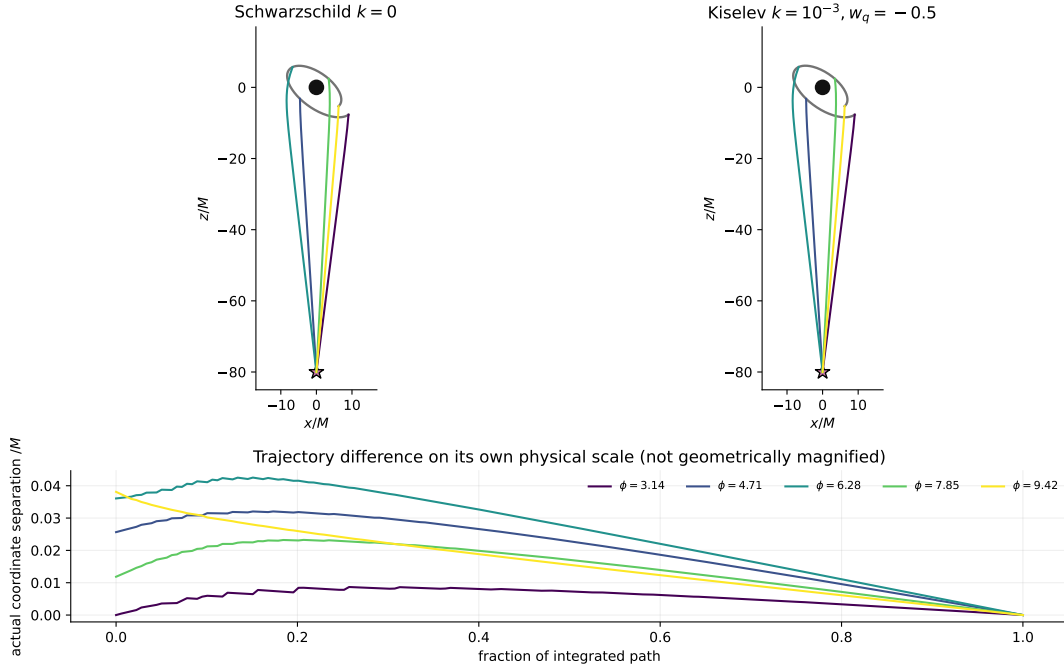


Figure 16: Matched-phase direct rays for Schwarzschild and $k = 10^{-3}$, $w_q = -0.5$ with $M = 1$, $r_p = 8M$, $r_a = 12M$, and observer $(0, 0, -80M)$. The lower panel gives physical coordinate separation versus integrated path fraction. Differences are shown on their actual scale and are not geometrically magnified.

J Uniform 161-Phase Convergence Audit

All 121 points complete in 4473.2 s with zero failed points. The audit preserves the exact $k = 0$ boundary and uses no mixed-resolution row, interpolation, or proxy substitution.

K Reproducibility Summary

The physical grid contains 121 points. The uniform 161-phase audit completed all points, and the targeted 321-phase audit completed all 35 preselected points. Exact $k = 0$ rank loss is preserved. Registered split assignments are reused, preprocessing is fitted on training data only, and the traditional inverse numerical audit passed. The independent physics audit passed with the launch-angle warning documented below.

L Independent Physical-Shooting Audit

The independent reference implementation imported no production `bhhairml` physics code and compared against archived production outputs rather than rerunning the full 321-phase campaign. Invariant launch directions, rays, observer-plane hits, physical observables, extracted features, and selected Jacobians agree. The verdict is `PASS_WITH_WARNING`: raw launch angles need not be unique because equivalent angular branches encode the same unit direction and physical ray. Raw angle equality is therefore not a required invariant, and the warning is not a physics discrepancy.

M Finite-Domain Observable-Ambiguity Audit

This audit evaluates all pairwise distances on the registered 121-point physical grid. It cannot rule out isolated or continuous degeneracies between sampled points and therefore does not establish continuous global injectivity. For points i, j , normalized parameter separation is

$$d_{\theta}(i, j) = \sqrt{\left(\frac{k_i - k_j}{\Delta k}\right)^2 + \left(\frac{w_{q,i} - w_{q,j}}{\Delta w_q}\right)^2}, \quad (6)$$

where both spans are taken from the registered physical domain. Observable differences use the archived grid-wide q95–q05 feature scales. The primary cross-set distance is the RMS of standardized feature differences rather than a raw Euclidean norm, because the combined set has 31 dimensions while ringdown has four.

At $k = 0$, the ringdown distance across all 55 nominal- w_q pairs is exactly zero. For ringdown plus photon geometry the minimum and median are zero and the maximum is 2.63×10^{-12} . This is the required positive control and reproduces the exact Schwarzschild structural degeneracy to numerical precision; these pairs are not counted as undesirable collisions.

Table 4: Closest finite- k RMS standardized observable pair at each minimum normalized parameter separation.

Feature set	$d_{\theta} \geq 0.10$	≥ 0.25	≥ 0.50	≥ 0.75
Ringdown	0.005832	0.012668	0.025170	0.041655
Rdown + photon geom.	0.007710	0.013406	0.031550	0.072741

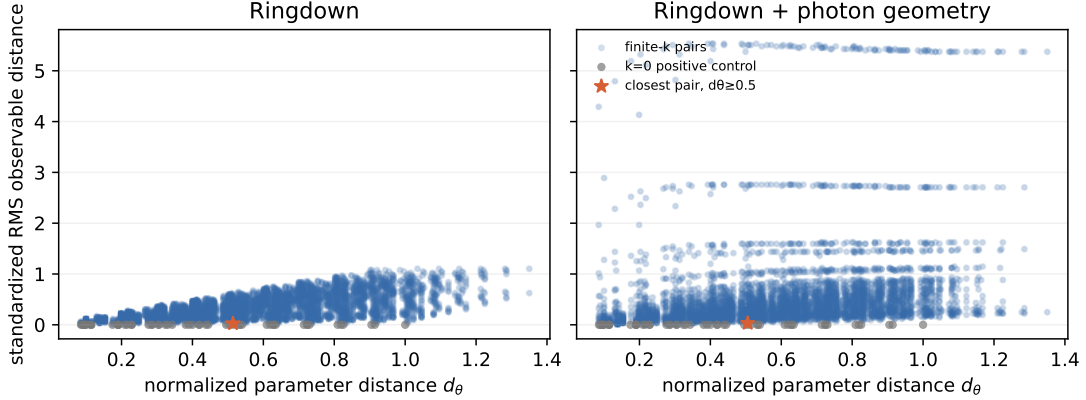


Figure 17: All-pairs finite-domain ambiguity audit. The horizontal axis is normalized parameter distance and the vertical axis is RMS standardized observable distance. Grey $k = 0$ pairs show the required structural degeneracy; distant finite- k pairs are the relevant sampled-domain diagnostic. Stars mark the closest finite- k pair with $d_\theta \geq 0.5$. This plot tests only the sampled physical grid and does not establish continuous global injectivity.

Table 5: Nearest-observable-neighbor parameter distance and preservation of the nearest parameter-space neighbor. Poorer exact local neighbor preservation does not imply increased distant ambiguity.

Feature set	median d_θ	p90	max.	top-1	top-3	top-5
Ringdown	0.09524	0.21513	0.23216	40.9%	71.8%	75.5%
Rdown + photon geom.	0.13171	0.21759	0.21886	20.0%	48.2%	67.3%

No nearest observable neighbor lies beyond $d_\theta = 0.25$ for either set, and no pair with $d_\theta \geq 0.25$ lies below the fifth-percentile local spacing. The closest combined pair at $d_\theta \geq 0.5$ is 12.8 times the largest measured combined-feature 161-to-321 RMS change. Photon geometry improves the distant-pair diagnostics but not every local ordering statistic.

N MLP Extrapolation-Failure Audit

The machine-readable table contains standardized input magnitude, hidden activation magnitude, iteration status, overflow flag, training distance, and resolution response for every catastrophic record. It distinguishes optimization failure from extrapolation instability without clipping.

Exact frozen-estimator shift summaries used in Fig. 12 are retained in Table 6 for numerical reference.

Table 6: Targeted 321-phase frozen-estimator robustness.

Model	median	95th percentile	max. w_q
HGB	0	0.08122	0.30593
RF	0	0.08509	0.27053
MLP	0.001143	0.008223	0.03819

References

- [1] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification, 2021.

- [2] Emanuele Berti, Vitor Cardoso, and Clifford M. Will. Gravitational-wave spectroscopy of massive black holes with the space interferometer lisa. *Physical Review D*, 73:064030, 2006.
- [3] Swetha Bhagwat, Xisco Jimenez Forteza, Paolo Pani, and Valeria Ferrari. Ringdown overtones, black hole spectroscopy, and no-hair theorem tests. *Physical Review D*, 101:044033, 2020.
- [4] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001.
- [5] Vitor Cardoso, Alex S. Miranda, Emanuele Berti, Helvi Witek, and Vilson T. Zanchin. Geodesic stability, lyapunov exponents and quasinormal modes. *Physical Review D*, 79:064016, 2009.
- [6] Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117:30055–30062, 2020.
- [7] Maximilian Dax, Stephen R. Green, Jonathan Gair, Jakob H. Macke, Alessandra Buonanno, and Bernhard Scholkopf. Real-time gravitational-wave science with neural posterior estimation. *Physical Review Letters*, 127:241103, 2021.
- [8] Olaf Dreyer, Bernard Kelly, Badri Krishnan, Lee Samuel Finn, David Garrison, and Ramon Lopez-Aleman. Black-hole spectroscopy: testing general relativity through gravitational-wave observations. *Classical and Quantum Gravity*, 21:787–804, 2004.
- [9] Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29:1189–1232, 2001.
- [10] Hunter Gabbard, Chris Messenger, Ik Siong Heng, Francesco Tonolini, and Roderick Murray-Smith. Bayesian parameter estimation using conditional variational autoencoders for gravitational-wave astronomy. *Nature Physics*, 18:112–117, 2022.
- [11] Stephen R. Green, Christine Simpson, and Jonathan Gair. Gravitational-wave parameter estimation with autoregressive neural network flows. *Physical Review D*, 102:104057, 2020.
- [12] V. V. Kiselev. Quintessence and black holes. *Classical and Quantum Gravity*, 20:1187–1198, 2003.
- [13] Pang Wei Koh, Shiori Sagawa, Henrik Marklund, et al. Wilds: A benchmark of in-the-wild distribution shifts. *Proceedings of the 38th International Conference on Machine Learning*, 139:5637–5664, 2021.
- [14] R. A. Konoplya and Z. Stuchlik. Are eikonal quasinormal modes linked to the unstable circular null geodesics? *Physics Letters B*, 771:597–602, 2017.
- [15] Yaniv Ovadia, Emily Fertig, Jie Ren, et al. Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift. *Advances in Neural Information Processing Systems*, 32, 2019.
- [16] Costantino Pacilio, Swetha Bhagwat, and Roberto Cotesta. Simulation-based inference of black hole ringdowns in the time domain. *Physical Review D*, 110(8):083010, 2024.
- [17] George Papamakarios, Eric Nalisnick, Danilo Jimenez Rezende, Shakir Mohamed, and Balaji Lakshminarayanan. Normalizing flows for probabilistic modeling and inference. *Journal of Machine Learning Research*, 22:1–64, 2021.

- [18] Fabian Pedregosa, Gael Varoquaux, Alexandre Gramfort, et al. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [19] Apostolos F. Psaros, Xuhui Meng, Zongren Zou, Ling Guo, and George Em Karniadakis. Uncertainty quantification in scientific machine learning: Methods, metrics, and comparisons. *Journal of Computational Physics*, 477:111902, 2023.
- [20] Andreas Raue, Clemens Kreutz, Thomas Maiwald, et al. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25:1923–1929, 2009.
- [21] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candes, and Aaditya Ramdas. Conformal prediction under covariate shift. *Advances in Neural Information Processing Systems*, 32, 2019.
- [22] Alejandro F. Villaverde, Antonio Barreiro, and Antonis Papachristodoulou. Structural identifiability of dynamic systems biology models. *PLOS Computational Biology*, 12:e1005153, 2016.
- [23] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- [24] Ariadna Uxue Palomino Ylla. Black-hole-hair machine-learning and geodesic-shooting software. <https://github.com/Uxuee/black-hole-hair-ml>, 2026.
- [25] Ariadna Uxue Palomino Ylla, Kosuke Makino, Akane Tanaka, Akihiro Ishibashi, and Chul-Moon Yoo. Ringdown waves from hairy black holes. *Journal of Cosmology and Astroparticle Physics*, 2026(09):046, 2026.